

QUCHAN, Iran, Islamic Republic of

WMO: 407400

Lat: 37.117N

Lon: 58.468E

Elev: 4222

StdP: 12.59

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.0	(c) 13.3	(d) 1.3	(e) 6.9	(f) 11.2	(g) 7.3	(h) 9.3	(i) 18.4	(j) 22.9	(k) 35.0	(l) 20.6	(m) 33.7	(n) 2.8	(o) 60	(p) 0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	94.1	62.8	91.5	61.9	89.0	61.0	67.2	84.0	65.5	83.0	64.0	82.0	10.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.8	96.5	74.4	59.6	89.3	73.4	57.7	83.2	71.9	34.2	83.7	32.8	83.1	31.5	81.6	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.5	21.3	18.4	DB	2.4	99.3	7.8	2.9	-3.3	101.4	-7.8	103.1	-12.2	104.8	-17.9	106.9	
			WB	0.7	70.1	8.5	1.8	-5.4	71.4	-10.4	72.5	-15.1	73.5	-21.3	74.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	31.6	36.1	45.8	55.0	64.2	72.1	75.8	73.3	67.1	56.4	44.1	36.0	(6)
	DBStd	16.68	9.13	8.90	8.71	7.88	5.63	4.38	4.14	4.30	5.29	6.85	7.71	7.26	(7)
	HDD50	1851	572	392	184	41	0	0	0	0	0	21	206	435	(8)
	HDD65	4678	1036	809	596	310	83	3	0	1	38	275	628	899	(9)
	CDD50	3637	1	3	54	189	442	664	799	722	514	220	28	1	(10)
	CDD65	991	0	0	1	8	60	218	334	258	103	9	0	0	(11)
	CDH74	12312	0	1	20	145	732	2683	4080	3164	1267	214	6	0	(12)
	CDH80	5028	0	0	2	23	187	1117	1923	1334	405	37	0	0	(13)
(14) Wind	WSAvg	6.7	4.8	6.1	7.4	7.9	7.7	8.5	7.7	6.7	6.9	6.4	5.8	4.9	(14)
(15) Precipitation	PrecAvg	11.3	1.3	1.4	2.1	1.7	1.3	0.6	0.2	0.1	0.2	0.4	1.2	1.0	(15)
	PrecMax	16.9	3.1	2.7	5.7	4.1	3.0	2.1	0.6	0.3	0.8	1.8	4.6	2.3	(16)
	PrecMin	4.3	0.3	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.8	0.7	0.7	1.0	1.1	0.8	0.6	0.2	0.1	0.2	0.4	0.9	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	64.3	76.6	83.4	88.8	95.6	98.0	96.4	92.4	83.8	73.2	60.8	(19)
		MCWB	44.7	47.1	52.4	57.3	61.0	63.5	64.5	62.4	59.6	56.8	52.6	45.3	(20)
	2%	DB	54.4	57.4	70.1	78.1	84.8	92.1	94.6	92.6	87.8	79.6	65.9	56.1	(21)
		MCWB	42.6	44.9	50.5	56.3	60.6	61.7	63.7	61.5	58.7	55.9	49.4	43.8	(22)
	5%	DB	49.3	53.5	65.8	74.1	81.5	89.2	92.1	89.6	84.6	75.6	61.8	52.1	(23)
		MCWB	40.4	43.1	49.5	54.6	59.5	61.0	63.0	60.4	58.2	54.4	47.7	42.5	(24)
	10%	DB	44.9	49.3	61.0	69.4	78.0	86.5	89.3	87.1	81.4	71.4	57.9	48.2	(25)
		MCWB	38.2	41.1	47.9	53.3	58.5	60.4	62.2	59.5	57.5	53.0	46.2	40.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	49.4	55.1	60.6	64.8	67.1	70.0	69.3	64.7	59.3	53.7	47.5	(27)
		MCDB	55.4	60.3	69.6	75.8	80.3	86.5	87.3	83.8	80.3	77.6	69.7	57.2	(28)
	2%	WB	43.5	46.3	52.3	58.0	62.9	65.2	67.9	66.3	62.7	57.3	51.3	45.0	(29)
		MCDB	52.5	54.6	65.7	73.0	78.7	84.4	85.4	81.7	79.1	74.6	62.9	53.4	(30)
	5%	WB	41.1	43.9	50.4	56.3	61.4	63.7	66.2	64.4	60.9	55.8	49.2	43.1	(31)
		MCDB	49.0	51.3	62.9	70.2	76.8	82.6	83.9	80.9	78.2	72.0	59.1	50.8	(32)
	10%	WB	38.2	41.8	48.6	54.6	60.0	62.4	64.6	62.3	59.3	54.2	47.1	40.7	(33)
		MCDB	44.4	48.7	60.2	66.6	74.7	80.9	83.0	80.7	77.0	68.6	55.9	47.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	18.9	21.2	21.6	24.3	27.0	26.5	27.8	28.2	26.1	21.2	19.1	(35)
		MCDBR	24.6	25.0	29.0	28.4	29.8	32.1	30.0	30.4	31.9	31.8	28.2	25.9	(36)
	5% WB	MCWBR	15.0	14.0	13.2	11.1	11.2	12.0	12.1	12.7	13.4	14.7	15.3	15.5	(37)
		MCWBR	22.8	21.3	25.7	24.5	26.5	26.7	25.4	24.4	27.1	28.2	23.3	22.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.353	0.429	0.476	0.492	0.472	0.472	0.424	0.390	0.402	0.355	0.312	(40)
	taud		2.305	2.177	1.968	1.900	1.894	1.950	1.954	2.078	2.150	2.082	2.196	2.321	(41)
	Ebn at Noon		274	275	262	257	254	259	258	268	271	253	253	266	(42)
	Edh at Noon		30	39	54	61	62	59	58	50	44	43	34	28	(43)
(44) All-Sky Solar Radiation	RadAvg		729	981	1328	1747	2127	2513	2489	2331	1914	1356	898	678	(44)
	RadStd		71	111	104	168	115	82	64	91	60	75	80	79	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	+0.92	-1.48	-1.44	N/A	N/A	N/A	+153